Relationship among Environment, Society and Technology

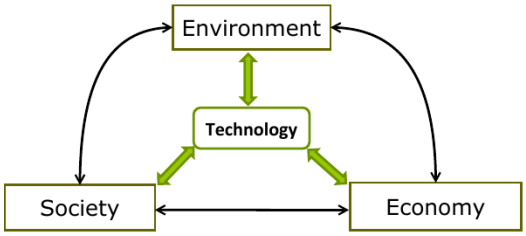
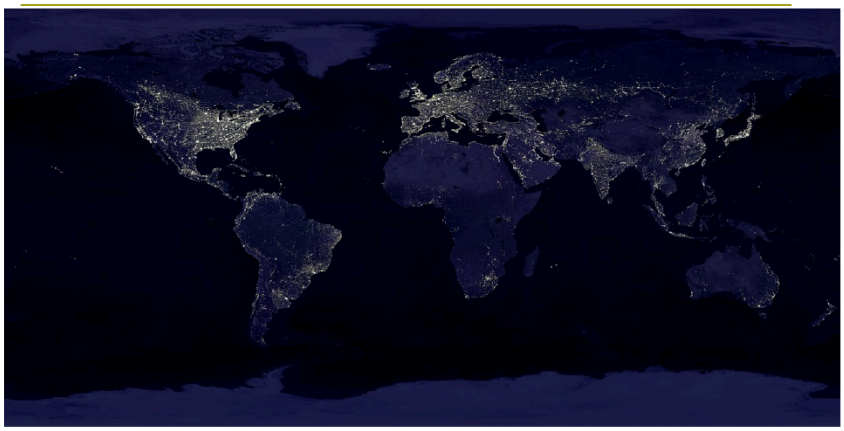


Figure: Dimension of Sustainable Triangle

Relationship among Environment, Society and Technology



NASA Image, 2012

Relationship among Environment, Society and Technology



Environment and Earth System

Do these three 'worlds' exist in total isolation from one another?

Of course not!!!

Interaction between PW, LW and HW

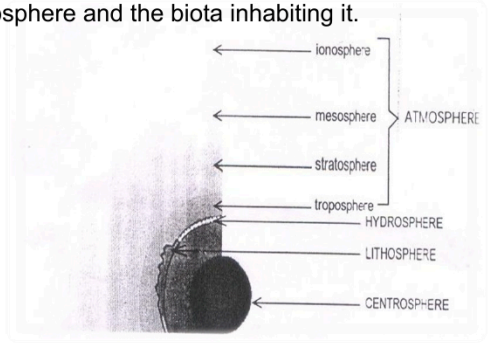
Inhale ⟶ Air (PW)
Eat ⟶ Food (LW)

http://www.yourarticlelibrary.com/essay/environment-and-its-components-explained-with-diagram/44682

Environment and **Earth System**

Earth System

The earth is a complex system consisting of different components; the lithosphere, the hydrosphere, the atmosphere and the biota inhabiting it.



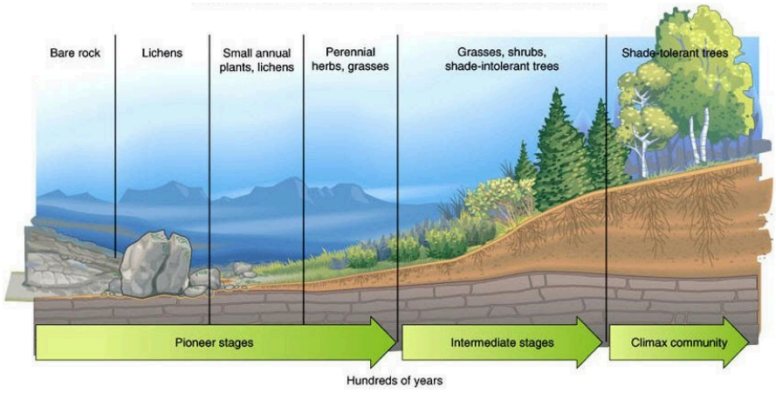
Environment and **Earth System**

Centrosphere
Earth was once a **molten mass** and to this date, the temperature and pressure in the interior of the earth is so high that the molten state still exists there. These are brought to the surface from time to time as **lava**, through **volcanic eruption** or through faults in the earth's crust.

Lithosphere
The lithosphere consists of the **solid crust** on which rest the continental masses comprising of **mountains, valleys and plains**. This is where human beings and other terrestrial animals live. The soil covering the solid masses in most cases is fertile enough to support plant life and in its natural state is covered with vegetation.

Ecology and Ecosystem

Ecological Succession

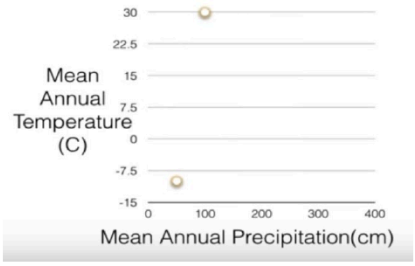


Ecology and Ecosystem

Therms:

- Mega: High, T.rain Forest
- Meso: High+Low, T. Deciduos Forest
- Micro: Low Temperature, Mixed Coniferous
- Hekisto: Extremely low, Alpine

Biomes: very large ecological areas; climate, vegetation, soil



Ecology and Ecosystem

A biome is determined mainly by its climate-like temperature and rainfall.
The earth's land areas are divided into **6 major biomes**.

1. TUNDRA

- Bitterly cold and covered with snow and ice.
- Ground remains permanently frozen (permafrost).
- Only certain small plants such as mosses and grasses can grow.



2. CONIFEROUS FOREST (TAIGA)

- Confers = cone-bearing trees (pines, fir)
- Cold climate
- Dense forests (tree tops block out a lot of sunlight)
- Grasses, small trees cannot grow –only some shrubs, ferns and mosses thrive.
- Animals such as squirrels, moose (sambur), birds, and insects live here.



Ecology and Ecosystem

Population

A group of organism of the **same species** which live in the same habitat at the same time where they can freely interbreed

The black-veined white butterfly *(Aporia crataegi)* mating



Ecology and Ecosystem

Biodiversity

The total number of different species in an ecosystem and their relative abundance





Ecology and Ecosystem

The characteristics of the type of environment where an organism normally lives. (e.g. a stoney stream, a deciduous temperate woodland, Bavarian beer mats)

Habitat



Ecology and **Ecosystem**

Types of Ecosystem: **on the basis of the size**

- Mega Ecosystem: Entire Biosphere
- Macro Ecosystem: Major Biomes like Tropical Rain forest (Amazon)
- Micro Ecosystem: Pond
- Nano Ecosystem: Drop of water

COMPONENTS OF ECOSYSTEM

1. **Biotic Components** includes living component from smallest microbe, fungi to largest animals and plants
2. **Abiotic Component** includes Inorganic substance (carbon, nitrogen, CO_2, H_2O etc.) involved in natural cycles, air, water and other physical factors, organic compounds (Proteins, Carbohydrates, Humus)

Ecology and **Ecosystem**

Among all abiotic components; The important ones are:

- **Temperature**; spatial distribution of species decreases on either side of equator towards the poles and as we go above and below the MSL.
- Water; Life started in water about 3.5 billion years ago.
- Light; photosynthesis
- Soil; Edaphic factors. Based on formation process, it is classified in two types:

Residual soil – Weathering and pedogenesis occur at the same place
Transported soil - Weathering and pedogenesis occur at different places (Colluvial- Brought by gravity, Alluvial- brought by water, Glacial- by snow, Eolian- by wind)

Structure and Function of Ecosystem

Vertical distribution species occupying different levels is called stratification. For instance, trees canopy is top layer of a forest while grasses are at the bottom.

Functions:
a) Productivity b) Decomposition c) Energy Flow and d) Nutrient Cycle

a) Productivity
Biomass: Mass of living biological organism in a given area at a given time

species biomass-mass of one species
community biomass- mass of all species in the community (including microorganisms, plants or animals). It is measured either in terms of dry weight (60-80% weight of water not included) or by measuring organically bound carbon.

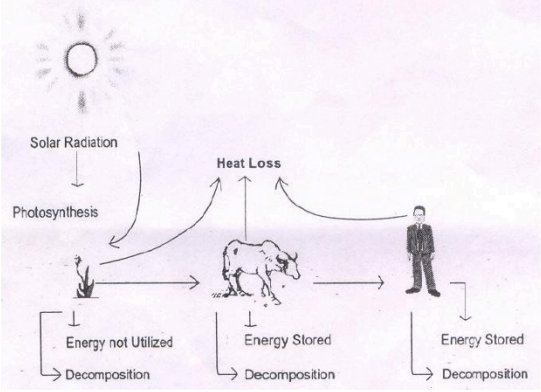
Structure and Function of Ecosystem

Rule of 10% (**10% law of Lindeman**): energy is reduced by 10% during flow to every higher trophic level. Because of this law, the number of trophic levels are usually restricted to 4-5.

Right from the sunlight being trapped by autotrophs into organic material, and further transfer of energy to higher trophic levels. On an avegrage, 10% (2-20%) of energy is transferred while rest 90% (80-98%) is lost in respiratory and other losses. So as the number of trophic levels increase in a food chain, most of the energy is lost.

Structure and Function of Ecosystem

Energy Flow in the Food Chain



Carrying Capacity

Carrying capacity of environment or an ecosystem is the **threshold limit** of use of that system without damaging it.

Every ecosystem has its resources that are used for economic development, for survival and for habitat creation. Environment and ecosystems have got the abilities to recover the loss of its resources by regenerating them over the period of time that are temporary and not exceeding the threshold damage limit.

